

UNIVERSITÄT LEIPZIG

World Models for Lenia Cellular Automata

Seminar Report — *Mathematics and AI Internship*

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1 Introduction

2 Data

2.1 Lenia Simulation and Data Generation

2.2 Dataset Versions

2.2.1 v1 — Random-noise Initialisation

2.2.2 v2 — Pre-placed Organisms

3 Methods

3.1 Model Architectures

We compare three world model architectures, all trained on the v2 Lenia dataset. Each model takes a single grayscale frame $\mathbf{x}_t \in [0, 1]^{64 \times 64}$ as input and produces a prediction $\hat{\mathbf{x}}_{t+1}$.

3.1.1 Pixel-CNN

3.1.2 Pixel-ViT

3.1.3 Patch-JEPA

3.2 Evaluation Setup

3.3 Intervention Experiments

A model trained purely on next-frame MSE could achieve deceptively strong rollout accuracy without learning any of the underlying cellular-automaton dynamics, since consecutive Lenia frames are highly autocorrelated and a near-identity prediction already scores well (Section ??). To disentangle genuine physical understanding from simple frame-copying, we designed a set of interventions that perturb the simulation state mid-rollout and test whether each model’s prediction still matches the behaviour of the true Lenia update rule. Concretely, each model is first rolled out autoregressively for $t^* = 10$ steps from a held-out initial frame, using only its own predictions as input. The resulting frame is then edited with one of five manually applied perturbations and fed back into the model for a further 30 steps. The identical perturbation, with matching random parameters (blob position, noise seed, etc.), is applied to the ground-truth Lenia simulation at the same step, so that both trajectories start from the same intervened state and can be compared directly at every subsequent step. We evaluate five interventions, each targeting a different aspect of the underlying physics. First, we inject a new "organism" (a Gaussian-blob seed) as intervention to test whether newly introduced structures are correctly integrated. A second intervention method is zeroing a circular region to test whether local death and regrowth are modelled. Furthermore, mirroring a small patch was applied to test sensitivity to local symmetry breaking. To test whether

global mass/intensity relationships are tracked we globally scaled the frame’s density. Lastly, we intervened by adding pixel-wise Gaussian noise to test robustness to low-level corruption. Since no model observed perturbed states during training, all five interventions are inherently out-of-distribution, so a correct continuation reflects generalised understanding of the Lenia update rule rather than memorised transitions. Results are averaged over five active (non-dying) validation trajectories, excluding trajectories in which the organism decays before the evaluation window ends, to avoid the trivially low error scores produced when every model simply predicts a black frame.

4 Results

5 Discussion

6 Conclusion